Mukesh Kumar

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Summary

- Data Scientist with 4+ years in BFSI, adept at turning intricate datasets into actionable insights. Experienced in predictive modeling, model validation, A/B testing, model optimization, forecasting and data visualization.
- Proficient in SQL, Python, R, Tableau and PySpark with strong foundation in statistics and ML from IIT Madras.
- Proven ability to lead teams, extract measurable business value from large-scale financial datasets, deliver scalable data-driven solutions, and communicate findings effectively to both technical and business audiences.

Professional Experiences

MathLogic | Gurugram, Haryana

Manager - Data Science & Analytics | Oct 2025 – Present

Consultant - Data Science & Analytics | Oct 2024 – Sep 2025

- Built custom credit scorecards for **New-to-Credit MSMEs** using bureau data and **proxy borrower behavior**, developing segment-specific **XGBoost** models by loan size (**KS up to ~34**) to enable risk-based pricing and expand credit access, leading to a **13% reduction in defaults** and higher approval rates for **thin-file MSMEs**.
- Developed a robust **GLM-based, industry-wide credit risk scorecard** for personal loans using **bureau features**, delivering a **13-point GINI lift over CIBIL with KS ~33**; cross-validated and adopted by multiple banks/NBFCs, reducing defaults, enhancing risk-return trade-offs, improving approval efficiency, and underwriting decisions.
- Built an **application scorecard** for unsecured business loans with **WOE binning & logistic regression**, reducing **1,329** features **to 11** key predictors and achieving **71% bad capture** in the **top 30% risk** band (**AUC 0.75, KS 35**).
- Led development of **1,800+ K-Means–based** peer group algorithms at a leading credit bureau, boosting legacy credit scoring performance by **20–23% (statistically significant, p < 0.05)**; owned full model lifecycle from data processing and feature engineering through **model validation**, governance reviews, and stakeholder delivery.
- Built a **Propensity-to-Pay scorecard** for DPD 90+ portfolios, capturing **~70% of recoverable NPA** accounts (≥5% repayment within 3 months) in the **top 30% scored** segment **(KS ~33)**, enabling focused recovery prioritization.
- Performed **EDA using SQL** and Excel on **100Cr+ credit records** to identify **risk anomalies and fraud patterns**; validated **700+ key-risk indicators** aligned with **RBI compliance**, such as identity theft and behavioral risk flags.

BioUrja Group | Indore, Madhya Pradesh

Data Scientist | Jan 2022 – Jun 2024

- Designed and deployed a **high-accuracy demand forecasting model** for U.S. natural gas (Python, Scikit-learn) with **R^2 > 90%**, quantifying weather and commodity sensitivities to inform **$50M+** portfolio risk decisions.
- Streamlined data pipelines and BI workflows across **100+ SQL** assets and **50+ Tableau dashboards**, boosting trading decisions by **55%** via long-term fundamentals and live market data integration, driving **$10M+** revenue.
- Delivered a **38% improvement in forecast** accuracy by upgrading **time-series models to XGBoost** and applying targeted **feature engineering** to reduce dimensionality by **98%**, enhancing driver-level explainability.

Technical Skills

- **Programming Languages & Tools:** SQL | Python (Pandas, Scikit - Learn) | R | Tableau |PySpark | Hadoop | GCP
- **Statistics & ML:** Regression | Regularization (L1/L2) | Bagging (Decision Trees, Random Forest) | Boosting (XGBoost, GBM) | Clustering (KMeans) | A/B Testing | Model Validation | EDA | KPI Tracking | Forecasting

Education & Scholastic Achievements

Master of Technology (M.Tech), Industrial Mathematics & Scientific Computing

Indian Institute of Technology Madras | Chennai, Tamil Nadu **|** Jul 2019 – Jul 2021

Courses: Statistics for Data Science, Data Analysis & Visualization, Machine Learning, Mathematical Modeling

Achievements: AIR 71 out of 53,995 in CSIR-NET Mathematics Dec 2019 **|** 95.10 percentile in GATE Math 2019